

Chapter Three

Crystal Binding

Main interaction responsible to hold the crystals together are the attractive electrostatic interaction between the negative charge of electrons and positive charge of nucleus are most important interaction. Other forces such as magnetic forces are weak force and gravitational forces are negligible effect.

The cohesive energy of a crystal is defined as the energy that must be added to the crystal to separate its components into neutral free atom at rest, at infinite separation with electronic configuration. The term lattice energy is used in the discussion of ionic crystal and is defined as energy that must be added to crystal to separate its component ions free ions at the rest at infinite separation.

3.1 Crystal of Inert Gases

The inert gases form simplest crystals. The electron distribution is very close to the free atoms. They are transparent insulator, weakly bound, with low melting temperatures. The atoms have high ionization energies.

The outer most electron shell of the atom are completely filled and distribution of electron charge in free atom is spherically symmetric. The electron distribution in the crystal cannot be significantly distorted from the electron distribution around the free atoms, because cohesive energy is less than ionization energy.

Van der Waal - London Interaction.

Consider two identical inert gas atoms at separation R comparable with radii of atoms. If the charge distribution on the atom were rigid, the interaction between atoms would be zero, because the electrostatic potential of spherical distribution of electronic charge cancelled outside a neutral atoms by the electrostatic potential of the charge on nucleus. Then inert gas atom show no cohesion and could not condense. But the atoms induce dipole moment in each other, and the induced moments cause an attractive interaction between atoms.



Figure 3.1: Coordinates of the two oscillators

As a model we consider two identical linear harmonic oscillator 1 and 2 separated by R each atom experience charges $\pm e$ with separation x_1 and x_2 , the particle oscillate along x axis. Let

p_1 and p_2 denote momenta and c is force constant.

$$H_0 = \frac{1}{2m} p_1^2 + \frac{1}{2} c x_1^2 + \frac{1}{2m} p_2^2 + \frac{1}{2} c x_2^2$$

Assume that $c = m\omega_0^2$

$$\text{Let the } H_1 = \frac{e^2}{R} + \frac{e^2}{R+x_1-x_2} - \frac{e^2}{R+x_1} - \frac{e^2}{R-x_2}$$

$$|x_1|, |x_2| \ll R \text{ we expand } H_1 \simeq -\frac{2e^2 x_1 \cdot x_2}{R^3}$$

$$x_s = \frac{1}{\sqrt{2}}(x_1 + x_2) \quad x_a = \frac{1}{\sqrt{2}}(x_1 - x_2)$$

$$x_1 = \frac{1}{\sqrt{2}}(x_s + x_a) \quad x_2 = \frac{1}{\sqrt{2}}(x_s - x_a)$$

$$p_1 = \frac{1}{\sqrt{2}}(p_s + p_a) \quad p_2 = \frac{1}{\sqrt{2}}(p_s - p_a)$$

$$H = \left[\frac{1}{2m} p_s^2 + \frac{1}{2} \left(c - \frac{2e^2}{R} \right) x_s^2 \right] + \left[\frac{1}{2m} p_a^2 + \frac{1}{2} \left(c - \frac{2e^2}{R} \right) x_a^2 \right]$$

The two frequency of coupled equation is given by

$$\omega = \left[\frac{\left(c \pm \frac{2e^2}{R^3} \right)}{m} \right]^{1/2} = \omega_0 \left[1 \pm \frac{1}{2} \left(\frac{2e^2}{cR^3} \right) - \frac{1}{8} \left(\frac{2e^2}{cR^3} \right)^2 \dots \right] \text{ with } \omega_0 = \left(\frac{c}{m} \right)^{1/2}$$

The zero-point energy is given as $\frac{1}{2} \hbar (\omega_a + \omega_s)$ because of the interaction the sum is lowered

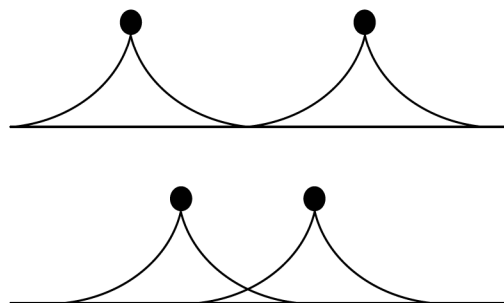
from the uncoupled value $2 \cdot \frac{1}{2} \hbar \omega_0$ by $\frac{1}{2} \hbar (\Delta \omega_s + \Delta \omega_a) = -\hbar \omega_0 \cdot \frac{1}{8} \left(\frac{2e^2}{cR^3} \right)^2 = -\frac{A}{R^6}$

The attractive interaction varies as the minus sixth power of the separation of two oscillators.

This is identified as Vander Waals interaction, it is also identified as London interaction or induced dipole -dipole interaction.

Repulsive term

As the two atoms are brought together their charge distribution gradually overlap, thereby changing the electrostatic energy of system. At sufficiently close separation the overlap energy is repulsive in large part because of Pauli exclusion principle. In the figure Electronic



charge distributions overlap as atoms approach. The solid circles denote the nuclei The experimental data on inert gas be fitted well by an empirical repulsive potential of the form

$\frac{B}{R^{12}}$, where B is positive constant.

Total potential energy and Lennard-Jones potential

Potential energy of Diatomic molecule in term of interatomic distance is given by

$U(R) = -\frac{A}{R^6} + \frac{B}{R^{12}}$ where $\frac{A}{R^6}$ is long range attractive potential and $\frac{B}{R^{12}}$ is short range repulsive potential.

Using data of Virial coefficient and viscosity new parameter ϵ and σ is introduced where $A = 4\epsilon\sigma^6$ and $B = 4\epsilon\sigma^{12}$ the Lennard-Jones potential is given by

$$U(R) = 4\epsilon \left[\left(\frac{\sigma}{R} \right)^{12} - \left(\frac{\sigma}{R} \right)^6 \right]$$

The force between two atom is given by $-\frac{dU}{dR}$

The value of σ and ϵ can be experimentally determined.

Generalization of theory

If Potential energy of Diatomic molecule is given by $U(R) = -\frac{A}{R^n} + \frac{B}{R^m}$ then force is given by

$$F(R) = -\frac{dU}{dR} \Big|_{R_0} \text{ which is equivalent to } F = -\left(\frac{nA}{R^{n+1}} - \frac{mB}{R^{m+1}} \right) = -\frac{nA}{R^{n+1}} + \frac{mB}{R^{m+1}}$$

For equilibrium separation at $R = R_0$,

$$F = 0 \Rightarrow \frac{nA}{R_0^{n+1}} - \frac{mB}{R_0^{m+1}} = 0 \Rightarrow \frac{nA}{R_0^{n+1}} = \frac{mB}{R_0^{m+1}} \Rightarrow R_0 = \left(\frac{nA}{mB} \right)^{\frac{1}{n-m}} \Rightarrow \frac{B}{A} \frac{R_0^n}{R_0^m} = \frac{n}{m}$$

The potential energy at equilibrium separation is given by

$$U(R_0) = -\frac{A}{R_0^n} + \frac{B}{R_0^m} \Rightarrow -\frac{A}{R_0^n} \left(1 - \frac{B}{A} \frac{R_0^n}{R_0^m} \right)$$

$$\text{Where } U(R_0) = -\frac{A}{R_0^n} \left(1 - \frac{n}{m} \right)$$

$$R_0 \text{ is stable equilibrium point so } \frac{d^2U}{dR^2} \Big|_{R=R_0} > 0 \Rightarrow m > n$$

Example 1: The potential energy of a system of two atoms is given by

$$U(R) = -\frac{\alpha}{R^4} + \frac{\beta}{R^{12}} \quad (1)$$

Calculate the amount of energy released when the atoms form a stable bond. Determine the bond length.

Solution: We know that the force of interaction between the two atoms is given by

$$F = -\frac{dU}{dR} = -\frac{4\alpha}{R^5} + \frac{12\beta}{R^{13}}$$

However, at an equilibrium distance (= bond length) $R = R_0$, the net force acting on the atoms is zero. Therefore,

$$F = -\left(\frac{dU}{dR}\right)_{R=R_0} = 0 \Rightarrow \frac{4\alpha}{R_0^5} = \frac{12\beta}{R_0^{13}} \Rightarrow R_0^8 = \frac{12\beta}{4\alpha} = \frac{3\beta}{\alpha}, R_0 = \left(\frac{3\beta}{\alpha}\right)^{1/8}$$

This is the required bond length.

Now substituting the value of R_0 in expression of potential energy.

Obtain the amount of energy released, i.e. $U_e = \frac{\alpha}{R_0^4} \left(1 - \frac{1}{3}\right) = \frac{\alpha}{\left(\frac{3\beta}{\alpha}\right)^{1/2}} \times \frac{2}{3} = -\left(\frac{4\alpha^3}{27\beta}\right)^{1/2}$

This is the required energy when the atoms form a stable bond.

Example 2: The potential energy of a system is given as $U(R) = -\frac{A}{R^6} + \frac{B}{R^{12}}$, the atoms form a stable bond length of $3A^0$ and bond energy of $1.8eV$. Calculate the force required to break the molecule and the critical interatomic distance for which it occurs.

Solution: $R_0 = 3A^0 = 3 \times 10^{-10} m$, $D = -U_0 = 1.8eV = 1.8 \times 1.6 \times 10^{-19} J$, $m = 12, n = 6$

$$D = -U = \frac{A}{R_0^n} \left(1 - \frac{n}{m}\right) \Rightarrow D = 2DR_0^6 = 2 \times 1.8 \times 1.6 \times 10^{-19} \times (3 \times 10^{-10})^6 = 4.19 \times 10^{-76} Jm^6$$

$$F = -\left(\frac{dU}{dR}\right)_{R=R_0} = -\frac{6A}{R_0^7} + \frac{12B}{R_0^{13}} = 0 \Rightarrow B = \frac{A(R_0)^6}{2} = \frac{A(R_0)^6}{2}$$

$$= \frac{4.19 \times 10^{-76} \times (3 \times 10^{-10})^6}{2} = 1.53 \times 10^{-133} Jm^{12}$$

The critical separation is given by $\frac{dF}{dR} = 0 \Rightarrow \frac{42A}{R_c^8} - \frac{156B}{R_c^{14}} = 0 \Rightarrow R_c^6 = \frac{13}{7} R_0^6 \Rightarrow R_c = \left(\frac{13}{7}\right)^{1/6} R_0$

$$R_c = 3.3A^0$$

Minimum force, $F_{\min} = -\left(\frac{dU}{dR}\right)_{R=R_c} = -\frac{6A}{R_c^7} + \frac{12B}{R_c^{13}} = -\frac{6A}{R_c^7} \left(1 - \frac{2B}{AR_c^6}\right) = -\frac{6A}{R_c^7} \left[1 - \left(\frac{R_0}{R_c}\right)^6\right]$

Where $\frac{B}{A} = \frac{R_0^6}{2}$ so $F_{\min} = -\frac{6 \times 4.19 \times 10^{-76}}{(3.33 \times 10^{-10})^7} \left[1 - \left(\frac{3}{3.33}\right)^6\right] = -2.58 \times 10^{-9} N$

Thus the force needed to dissociate the molecule is $2.58 \times 10^{-9} N$